

ISSN Print: 2617-4693
ISSN Online: 2617-4707
NAAS Rating (2025): 5.29
IJABR 2025; SP-9(12): 820-828
www.biochemjournal.com
Received: 18-09-2025
Accepted: 23-10-2025

Bikram Pradhan
Lecturer, Department of Biology, Bharat Institute of Science and Technology, Bhubaneswar, Odisha, India

Dr. Stephan J
Assistant Professor, REVA University, Bangalore, Karnataka, India

Amrita Kumari
Assistant Professor, Dr. C V Raman University, Bihar, India

Mani Prabhat
Research Scholar SHUATS, Prayagraj, Uttar Pradesh, India

Dr. V Narendhiran
Assistant Professor, Dhanalakshmi Srinivasan University, Tiruchirappalli, Tamil Nadu, India

Nikhil AN
PhD Scholar (Fishing Technology), Department of Fishing Technology, College of Fisheries, Mangaluru, Karnataka, India

Amit Kumar Patel
Assistant Professor, Dr. C V Raman University, Bihar, India

Corresponding Author:
Amit Kumar Patel
Assistant Professor, Dr. C V Raman University, Bihar, India

A comprehensive literature study of morphology, taxonomy, phylogeny and biochemical aspects of genus *Utricularia*

Bikram Pradhan, Stephan J, Amrita Kumari, Mani Prabhat, V Narendhiran, Nikhil AN and Amit Kumar Patel

DOI: https://www.doi.org/10.33545/26174693.2025.v9.i12Sj.6620

Abstract
Utricularia L. (bladderworts) is one of the largest known carnivorous plant genera which come under Lentibulariaceae family. It comprises of about 240 species that are classified into three subgenera and 32 sections. *Utricularia* have unique morphology and can colonize in wide range of ecosystems such as lithophytes, epiphytes, rheophytes, etc. These plants are excellent specimens for future researches on physiology of carnivorous plants, trap evolution, biochemistry of pray capture, microbial symbiosis, aquatic phytoremediation and molecular phylogeny of angiosperm genome. Recent studies have shown that, some species of *Utricularia* and *Genlisea* have the smallest genome size in angiosperms. Three lineages in *Utricularia* genus have evolved independently which have genome size of less than 100 Mbp. Many researchers have also reported that submerged aquatic macrophytes of *Utricularia* can accumulate heavy metals at higher concentrations. The present review aimed to study the taxonomy, molecular phylogeny of *Utricularia* and its utilization for phytoremediation of polluted water bodies.

Keywords: *Utricularia*, phylogeny, phytoremediation, bladderworts, lentibulariaceae

1. Introduction
Earth's ecosystem is full with organisms that are very mysterious and hold many surprises. Few of these organisms are carnivorous plants which are widely distributed all around the world. Although the carnivorous plants are present around us since thousands of years, the first descriptions of these plants were reported in fifteenth century by Auslasser (1479) on *Pinguicula* [Lonning and Becker, 2004] [37]. More than500 species of carnivorous plants have been reported and are classified into different families since then.
Utricularia L., commonly called as bladderworts is one of the largest known carnivorous plant genera which come under Lentibulariaceae family of order Lamiales. It comprises of about 240 species that are classified into three subgenra and 32sections [Taylor, 1989; Müller *et al.*, 2006; Albert *et al.*, 2010; Fleischmann, 2012; Fleischmann, 2015] [69, 43, 3, 18, 17]. All the *Utricularia* species are classified on the basis of general trap morphology. But bladderwort traps have complex physiology which is not fully understand yet [Poppinga *et al.*, 2016] [53]. It shows high phonotypical plasticity (Inter as well as Intra specific) which makes it very difficult to identify all species. Mostly these plants get nutrition from minute preys like small crustaceans but some of the *Utricularia* species like *Utricularia vulgaris* L. can feed on larger preys which include nematodes, mosquito larva, etc [Harms and Johansson, 2000; Mette *et al.*, 2000] [41].
Utricularia was first described by, de Lobel during sixteenth century in the year 1591. And in India, *Utricularia* came into limelight with the publication of Nelipu by van Rheedein 1689. Myanmarn (1768) reported about *U. bifida* for the first time along with *U. caerulea*. In the year 1804, 34 species of *Utricularia* was reported by Vahl which included five species described from India. In 1859, Monograph of Indian *Utricularia* was published by Oliver. He reported 24 species, including four new species, two new varieties and a new name- *U. kumaonensis* Oliver. In 1868, the first discovery of the trap functions of *Utricularia* was reported by Holland.

In 1884, Claeke reported 22 species from different parts of India, Pakistan, Nepal, Bhutan, Sri Lanka, Malaya and Myanmar. In 1922, Haines listed nine species from Bihar and Odisha. Peter Taylor in 1989 published The Genus *Utricularia*- A Taxonomic Monograph which included 33 species from India [Taylor, 1989] [69]. In 1992, M.K. Janarthanam and A.N. Henry published Bladderworts of India in Flora of India, Series 4 which contains morphology and taxonomic keys of *Utricularia*.

2. Methodology

Using carefully chosen keywords and Boolean operators, a thorough literature search was carried out across several databases (including Google Scholar, PubMed, Scopus, and Web of Science). This stage guarantees that every pertinent study is recorded. After that, the retrieved articles are checked in two steps: first, titles and abstracts are examined, and then the whole text is evaluated to verify eligibility. To prevent duplication and guarantee correctness, the chosen studies are arranged methodically, frequently with reference management software. Key information from each study, such as authorship, publication year, study design, primary findings, and conclusions, is gathered through data extraction. After that, the collected data is either statistically (for quantitative or meta-analytic reviews) or narratively (for qualitative reviews) examined and synthesized. In order to evaluate the studies' validity, reliability, and any biases, a critical evaluation is also carried out.

A standardized structure called Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) was created to guarantee repeatability, accuracy, and transparency in systematic reviews and meta-analyses [Page *et al*., 2021] [46]. It entails a number of crucial processes, starting with the formulation of a precise research topic and the creation of a thorough protocol that specifies inclusion and exclusion criteria, databases to be searched, and analysis techniques.

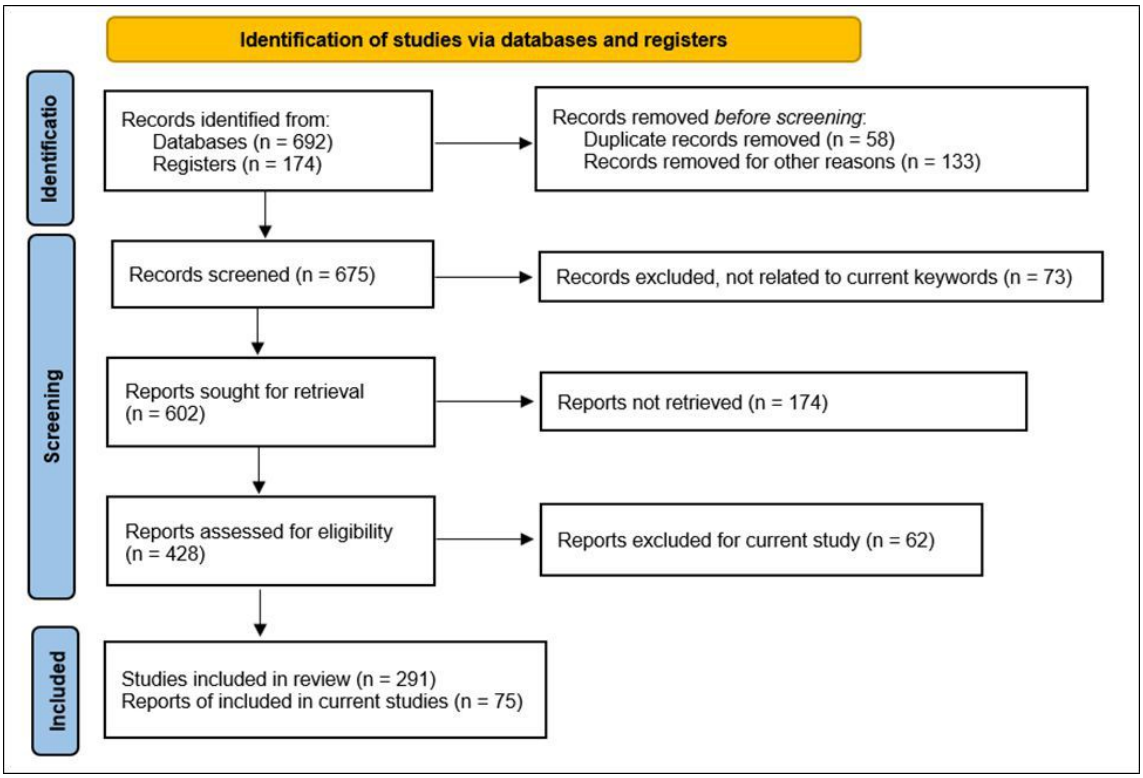


Fig 1: PRISMA flow chart which represent the systematic literature meta-analysis conducted for the current topic by using specific keywords

3. Results

In contrast to most plants, *Utricularia* uses its leaves and traps to directly absorb nutrients because it lacks actual roots. By pumping out water, the tiny bladder traps produce a vacuum. When tiny aquatic animals, such protozoa or small crustaceans, activate the trapdoor hairs, the door opens quickly, drawing in the prey. After the trap shuts, the prey is broken down by digestive enzymes, which enables the plant to absorb nutrients like phosphorus and nitrogen that are frequently lacking in their natural environments. *Utricularia* has pharmacological and ethnobotanical significance. Certain species have been used in traditional medicine to treat skin diseases, fevers, and respiratory and urinary conditions. Bioactive substances such flavonoids, terpenoids, and phenolic acids have been found in recent phytochemical research, indicating possible antibacterial, antioxidant, and anti-inflammatory qualities.

3a. Taxonomy of *Utricularia*

Utricularia is a very large and species-rich genus, and many species are often difficult to distinguish because of its unique morphology. A lot of different species and names have been published for over 200 years since Linnaeus put up the genus in 1753. The book published by Taylor, The Genus *Utricularia* is considered as base for the taxonomy of *Utricularia*. During his work, he came across more than 900 names for taxa of *Utricularia* that had been published. He studied and compared type specimens and descriptions, and finally was able to recognize that more than 3/4 of the names were actually representing synonyms of already described taxa, or invalid names [Fleischmann, 2012] [18].

Now a day's molecular phylogeny makes it easy to categories or identification of existing as well as newly discovered species [Jobson *et al*., 2003; Silva *et al*., 2018] [30, 66].

3. b. Morphology of *Utricularia*

Utricularia are herbaceous plants which cannot be classified on the basis of traditional methods based on morphological difference [Yadav *et al*., 2005] [75]. The vegetative parts of bladderworts are usually remain submerged in water or mud and due to this difference in habitat, variations in their morphology can be seen. Rutishauser and Isler (2001) stated that *Utricularia* do not have typical plant organs like stem, leaf or roots.

3. b. I. Rhizoids

Utricularia species do not have true roots, instead they have root like organs called rhizoids. They help in anchoring of the plant. In case of aquatic species rhizoids are not often present. Some species like *U. exoleta* have thick rhizoids where as some have branched rhizoids (*U. australis*). Branched rhizoids are mostly simple and papillose. In *U. aurea* the rhizoids are long and fusiform at the ends. In terrestrial and epiphytes, rhizoids are well developed at space base.

3. b. II. Stolon

Stolon is the pseudo-stem like structure present in *Utricularia* plants. Stolons in aquatic species remain submerged under water or free floating and can grow up to one meter in length. In case of terrestrial species, the stolons can be single or branched and are thread like (filiform) in appearance. Stolons in most of the species are soft and can be easily detachable. This makes it difficult to collects many species with intact stolons for taxonomical studies.

3. b. III. Foliar organs

True leaves are also absent in *Utricularia*. The shape, structure and venation of foliar organs vary from species to species. For example, in *U. aurea* the foliar organs range from 3-5 segments [Rahman, 2005] [55]. *U. minor* has polymorphic foliar organs.

3. b. IV. Inflorescence

Flowers (inflorescence) of *Utricularia* species are racemes and attached to the stolon. The size, shape and color of flowers differ species to species. In terrestrial species like *U. reticulata* or *U. scandens*, the flowers are twining. In species like *U. aurea* the inflorescence were attached to inflated rhizoids [Kumar *et al.*, 2018] [33]. The inflorescences of *Utricularia* are compared with beautiful orchids. The inflorescences of aquatic species like *U. vulgaris* are similar to small yellow snapdragons.

3. b. V. Traps

All carnivorous plant have traps to captured small preys. But the traps of *Utricularia* are very complex and also the fastest trapping device in the whole plant kingdom [Poppinga *et al.*, 2013; Poppinga *et al*., 2016] [52, 53]. The traps, also called as utricles, bladders or vesicular, are leaflike organs attached to the vegetative parts of the plants like stolon, rhiozoids and foliar organs [Janarthanam and Henry, 1992; Krol *et al*., 2012] [25, 31]. In case of epiphytes the traps are attached to stolon or rhiozoids. The opening (mouth) have trapdoors with several hair like structure (triggers) that makes the bladders watertight [Poppinga *et al.*, 2017] [49]. The walls of utricles have glands that pumps out the water, maintaining a negative hydrostatic pressure [Adamec, 2011; Sasago and Sibaoka, 1985a; 1985b; Vincent *et al*., 2011] [1, 62, 63, 72]. Due to this pressure the trap walls deflate and store elastic/potential energy like a spring. When a prey touches any trigger hair, the tarp inflates spontaneously, opening the door rapidly (>2 ms) [Poppinga *et al.*, 2017] [49].

Once the traps inflates, the water inside is pumped out again and the bladders are ready for next capture in about 10-15 minutes. The prey inside dies due to anoxia and get digested by hydrolytic enzymes [Adamec *et al*., 2010] [2]. According to many studies, a little is known about the diversity of traps and the effectiveness of prey capturing mechanism. In a broader prospective, such knowledge would indeed be essentially.

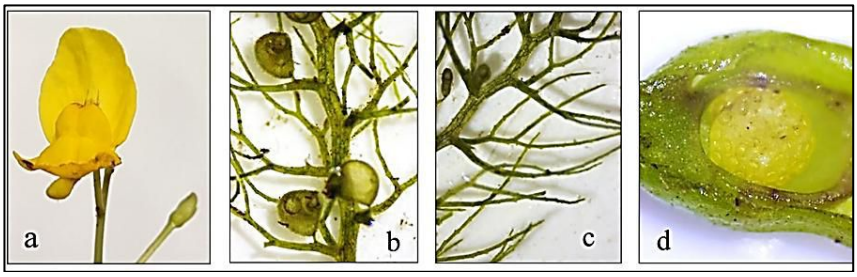


Fig 2: Morphology of *Utricularia aurea*: **a**. flower, **b**. traps, **c**. foliar organs, **d**. capsule and seeds.

3. c. Propagation in *Utricularia*

Most of the species *Utricularia* produced tiny dust like seeds which have small embryo [Płachno and Świątek, 2010; Eriksson and Kainulainen, 2011; Menezes *et al*., 2014] [47, 16, 40]. These seeds are dispersed by wind, insects, or by water. Capsule or fruit plays an important role in classification of species. In *U. aurea* the capsule are globose and circumscissile.Seeds also show great variation in *Utricularia*. Aquatic species like *U. aurea, U. minor and U. stellaris* are winged. Some species also form winter buds by modifying the shoot apices and leaves [Płachno *et al.*, 2014] [48]. These buds contain high protein and starch. Stolon fragments can also help in spreading of certain *Utricularia* species.

3. d. Interrelationship among the genera of the family Lentibulariaceae

Lentibulariaceae family comprises of above 300 species of carnivorous plants which are divided into three genera; *Utricularia*, *Genlisea* and *Pinguicula*. All the three genera are interrelated to each other on both morphological and molecular basis [Jobson*et* 2003]. *Utricularia* and *Genlisea*

have evolved drastically with body orientation completely differing from classic root-shoot rules in angiosperms [Rutishauser, 2016] [61]. Where as in *Pinguicula* species, normal vegetative parts like shoot, root and leaves are present [Roccia *et al.*, 2016] [59]. *Pinguicula* plants have flypaper like traps which are covered with glandular trichomes. In case of *Genlisea*, specialized Y shaped organs called rhizophylls are present which helps in plant anchoring and traps small organisms. Great variations are seen in morphology of traps in all the species of *Utricularia*. The bladders (traps) are hollow, 1-6 mm long with trapdoors at the entrance [Vincent *et al*., 2011] [72].

3. e. Distribution and habitat of *Utricularia*

Except the Polar Regions, *Utricularia* grows all around the world. *It* can be found mostly in tropical and sub-tropical region with highest taxa concentrations in America, Australia and Asia [Taylor 1989; Jobson *et al*., 2003; Müller and Borsch 2005; Guisande *et al.,* 2007; McPherson, 2010] [69, 30, 42, 22, 39]. South America and Australia are regions with high taxa concentration (76% of Australia and 65%of South America). The highest region is Africa with more than 40 species. In India these species were found in different parts of Himalayan ranges, Western Ghats and Eastern Ghats. *Utricularia* mostly grows in warm climate with temperature ranging from 25º- 35º Celsius. But few species can grow in low temperature as well as high temperature. *Utricularia* grow on wet surfaces near water bodies or in nutrient poor pounds (free floating aquatic species), or in habitat with more or less permanent or seasonal high humidity. For example, *Utricularia vulgaris*, grows in meso-eutrophic waters (pH between 5 and 8) of the northern hemisphere and 'in both Africa and India some *Utricularia* species are commonly found in rice cultivations' [Taylor, 1989; Lonnig and Becker, 2004] [69, 37]. These plants may even be considered as invaders of a more or less artificial no oligotrophic environment. Species which grown on rocks have thin layer of water surrounding the bladders. On the basis of the habitat, *Utricularia* is differentiated into six different groups.

Species that grows on wet soil near water bodies like pond or streams comes under this group. Rhizoids (organs looking like roots) are well developed and able to fix the plants to the soil surface, but unable to absorb water and nutrients. Filiform stolons are developing; they exceed rarely a few centimeters long. Both stolons and rhizoids contribute to anchor the plant. The stolons may bear scatterred leaves (*U. arenaria*) or clumped leaves (*U. simulans*). Species which can grow on still water comes under this group. Free floating species mostly remain submerged under water. The only visible part of the plant is the inflorescence. Vegetative organs are squashy and include no, or nearly so, lignified tissues (sclerenchyma). The collenchyma is, too, scarce or absent [Roques and Jérémie, 2005] [60]. Air shoot are present which help sin keeping the plant suspended in water.

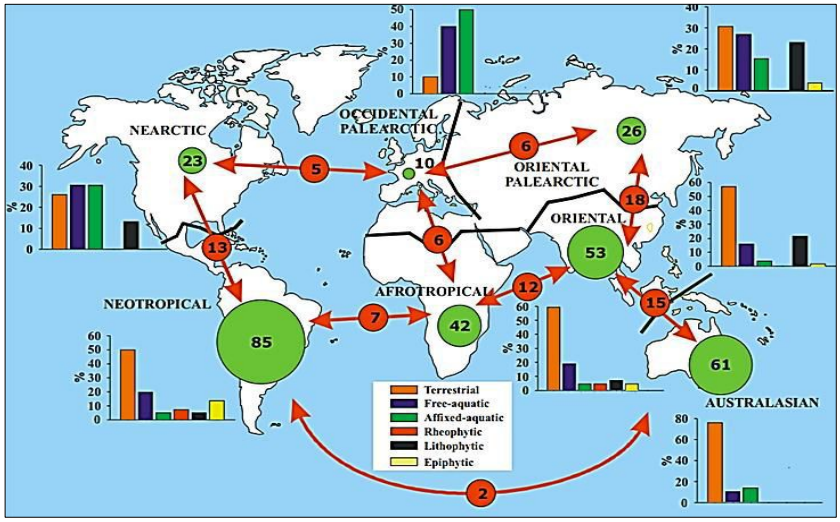


Fig 3: Biogeography of Utricularia. (Data collected from the research paper published by Guisande *et al*., 2007) [22]

Species which grows on still water but for better development fix itself to soil surface comes under this group. There are more than 50 such species whose trap-bearing stolon or rhizoids are anchored in thes oil, at bottom of the water (example, *U. minor*). Some species like *U. gibba*, though being floating plants, develop both specialized anchored parts while some others like *U. benjaminiana, U. rejlexa* produce tubers [Roques and Jérémie, 2005] [60]. Species which grows in swiftly moving water sources by anchoring onto a soil surface (*U rigida, U. tetraloba, U neottioides)*. Stolons and dissected leaves are reduced to filaments undulating in running water. Traps are rarely observed or not properly developed. Species that grows on moss covered trees or banks comes under this group. *U. jamesoniana, U. alpina, U. manni*iare examples of epiphytes which occurs in rain forests of very wet tropical climates. Species which grows on rocks comes under this group. Small ground species (i.e. *U. arenaria, U. prehensilis*) are found in shallow cavities, and ubiquitous semi-floating species (*U. gibba*sub sp. *exoleta*) in somewhat deeper hollows or cracks in therock.

3. f. Molecular phylogenetics and evolution

Recent studies had shown that some species of *Utricularia* and *Genlisea* have the smallest genome size in angiosperms [Ibarra-Laclette *et al*., 2013] [24]. The size of chromosome in these species is nearly equal to size of bacterial genome [Fleischmann *et al*., 2014; Greilhuber *et al*., 2006] [19, 21]. Three lineages in *Utricularia* genus have evolved independently which have genome size of less than 100 Mbp [Veleba *et al*., 2014] [70]. Therefore, these plants can be consider as extraordinary specimens to analysis the genome

miniaturization process. In some studies, different molecular markers such as *matK*, *rps16*, *trnk* are used for conducting phylogenic analysis in *Utricularia* [Jobson and Albert, 2002; Jobson *et al*., 2003; Reut and Jobson, 2010; Westermeier *et al*., 2017] [26, 30, 58, 74]. Silva and his coworkers (2018) have elaborated the molecular phylogeny of bladderworts (*Utricularia*) in a wider approach.They conducted molecular analysis by using different plastid genome like *rbcL*, *matK*, *rps16*, etc and observed that the *rbcL* gene shows very low variable sites whose conservation is as essential as Rubisco enzyme that carries out dark reaction in photosynthetic plants. Raven, 2013 and Shivakumar *et al*., 2016 [64] reported similar observation.

They found that subgenus *Polypompholyx* is the sister group to the clade formed by subgenus *Utricularia* and *Bivalvaria* according to the combined sequences and also individual *matK*, *rps16* intron and *trnL*-F datasets. Previous studies based on *trnL*-F spacer, *rps16* intron [Jobson and Albert, 2002; Jobson *et al*., 2003; Reut and Jobson, 2010] [26, 30, 58], and on *matK* [Silva *et al*., 2016] [65] also support this tree topology (*Polypompholyx, Bivalvaria, Utricularia*); although Müller and Borsch (2005) had placed subgenus *Polypompholyx* as a sister group to subgenus *Bivalvaria*, and both as sister group to subgenus *Utricularia*. By taking pollen into consideration, Lobreau-Callen *et al*., (1999) [36] recognized shared characters of both sections and Taylor (1989) [69].

3. f .I. Phylogeny of Trap diversity

Until now, there are few reports available which provide a detail or brief knowledge on behavior of prey in the proximity of motile or non-motile carnivores plants trap during trapping [Bohn and Federle, 2004; Bauer *et al*., 2012; Poppinga *et al*., 2012] [10, 8, 51]. During recent studies, analyses of trap kinematics were performed with artificially triggering traps [Forterre *et al*., 2005; Poppingaand Joyeux, 2011; Volkov *et al*., 2014] [20, 50, 73]. All this leads to many unsolved questions regarding it phenotypic and genotypic diversity as well as trapping mechanism. Westermeier (2017) [74] in his study 'Trap diversity and character evolution in carnivorous bladderworts (*Utricularia*, Lentibulariaceae)' reported about functional morphology and trapdoor movement based on phylogenetic reconstruction of genes, derived from the rapidly evolving chloroplast regions *trnK*, *rps16*, and *trnQ-rps16*.

3. f. II. Evolution of *Utricularia*

Lbarra- Laclettle *et al*., (2013) [24] stated about molecular clock on the basis of *trnL*, *rps16*, *coxl* and *maK* sequencing. Based on molecular clock it was calculated that about 42 million years ago the common ancestor *Utricularia*, *Genlisea* and *Pinguicula* got diverged [Bell *et al*., 2010] [9]. The last common ancestor of *Genlisea-Utricularia* clade was found to bearose 39 mya (36 mya estimated by Ibarra-Laclette *et al*., 2013) [24], a lineage from South America. About 30 mya (31 mya according to Ibarra-Laclette *et al*., 2013) [24] *Utricularia* got separated from its sister genus and dispersed to Australia with the lineage represented by subgenus *Polypompholyx* (17 mya) and possibly to Africa afterwards (16 mya).The dispersal *Utricularia* to North America region may possibly happened from South America, around 11 mya (Ibarra-Laclette *et al*., 2013) [24] and to middle of Miocene between 12 mya [O'Dea *et al*., 2016] [45].

The colonization of *Utricularia* into Eurasia probably occurred via the Bering Strait to Northwestern Asia and also to Europe at around 4.7 mya [De Chaine, 2008] [13]. The lineage of *U. aurea, U. inflexa* and *U. reflexa* possibly reached the African continentfrom Tropical Asia very recently, around 3 mya, which would bestrongly supported if *U. cornelianawas* indeed *U. reflexa* [Jobson, 2012; Fleischmann, 2015] [28, 17].

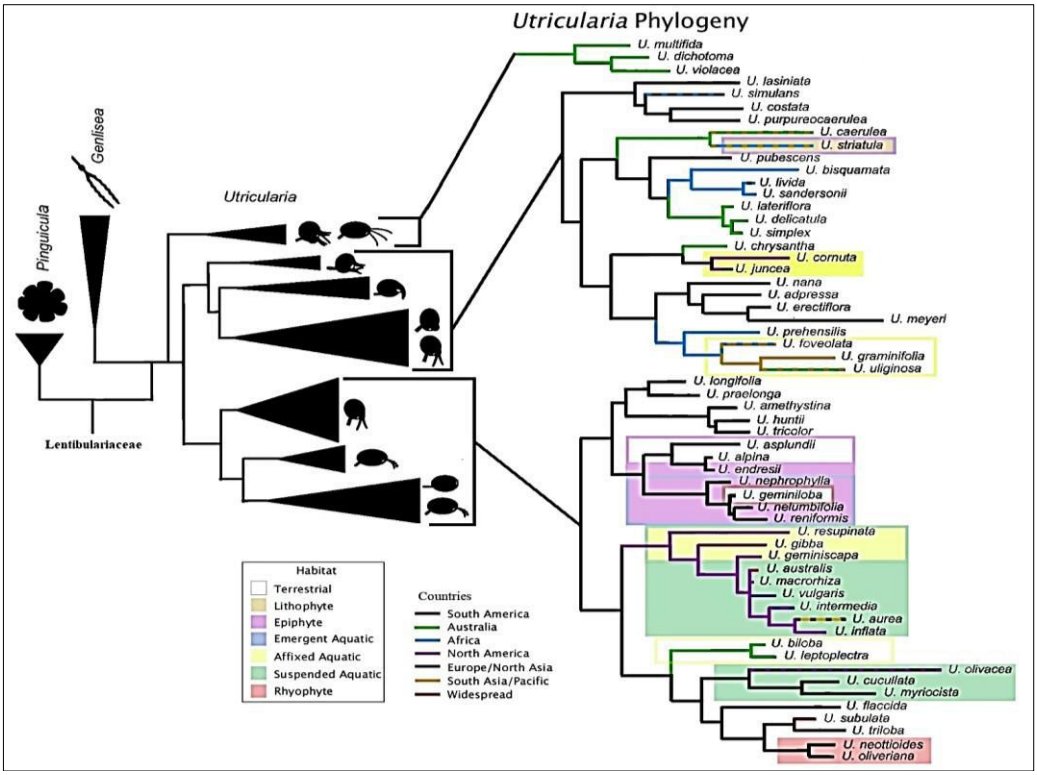


Fig 4: Pictorial representation of phylogeny linkage among different species of Utricularia along with the countries in which the species were reported as well as the habitats in which they grow by collecting the data from Victor *et al*., 2010; Jobson *et al*., 2003 [30] and Silva *et al*., 2018 [66].

3. g. Identification of new species

With the advancement in technologies and application of molecular phylogeny many new species of *Utricularia* are been discovered around the world. Yadav*et al*., (2005) [75] had identified a new *Utricularia* species (*Utricularia*babui) in the Western Ghats of India. In 2012, Fleischmen had presented many new species in the research paper "The new *Utricularia* species described since Peter Taylor's monograph". Jobson (2013, 2018) [27, 29] had identified five new species from Australia. In 2018, another new species of *Utricularia* (*Utricularia*sunilii) was found in Kerala, India.

3. h. Phytoremediation by *Utricularia*

Environmental pollution due to heavy metal pollutants has become a serious issue of modern society which have many hazardous effects [Prasad *et al*., 2006; Liu *et al*., 2007; Rai, 2009; Nouri and Haddioui, 2016] [54, 35, 56, 44]. Plants growing in heavy metal pollution have a spectrum of morphological and cellular changes, including reduced metabolism, degradation of biomolecules, change in proteins. These effects are caused due to stress by bioaccumulation of pollutants through root, stem or sprouts of plants growing in the polluted environment [Borišev *et al*., 2008; Babović *et al*., 2010] [11, 7].

Phytoremediation is the process of removing or detoxifying polluted water, soil and air by using plants. It has been proved by many researchers that plants can transfer pollutants into less harmful form as well as can accumulate metal pollutants in large quantities [Ali *et al*., 2013] [4]. Many aquatic species of plants can accumulate heavy metals without having any significant changes in their metabolism. Macrophytes and aquatic plants are often used for bioremediation of polluted water bodies [Borišev *et al*., 2008; Babović *et al*., 2010; Augustynowicz *et al*., 2010; Malec *et al*., 2011] [11, 7, 5, 38]. Submerged aquatic species of *Utricularia* such as *Utricularia vulgaris* and *Utricularia aurea* (macrophytes) can grow in heavy metal polluted water by phytoaccumilation.

Augustynowicz and his coworkers (2015) studied the bioremediation of chromium (VI) by *Utricularia gibba*. They observed that *Utricularia gibba* can accumulate Cr more effectively in comparison to other aquatic macrophytes like *Pistia stratiotes or Spirodelapolyrhiza*. *Utricularia aurea* helps in removal of heavy metal like zinc by biosorption method. Maksimovic (2019) [68] reported that *Salvinianatans* and *Utricularia vulgaris* have high bioaccumulation capacity to Fe, Zn, Mn, and Cu. The presence of common pioneer species (*Utricularia aurea* and *U. bifida*) indicates past disturbance in a habitat, such as that demonstrated in a paleonological study, with the first appearance of *U. aurea* pollen coinciding with the arrival of aborigines in Tasik Bera, Pahang. The presence of common *Utricularia* species in the suburban waysides can therefore be used as a rough prediction for water trophicity, succession stages, and the general health of the regenerating patches.

3. i. Medicinal properties of *Utricularia*

Although very little research has been conducted on medicinal values, it is still used traditional medicines. In Peninsular Malaysia, *Utricularia bifida* is used for treatment of urinary diseases. Divakar *et al*., (2013) [15] reported that Madayipara hillock tribals of Kerala use the *Utricularia reticulata* against urinary problems. In the year 2015, Deka and Devi reported that *Utricularia aurea* is used to cure diseases caused by mosquito. Choosawad *et al*., (2005) [12] reported the antioxidant activity of *Utricularia aurea*. Few research works have been conducted on medicinal properties of *Utricularia* species [Kumar *et al*., 2017; Singh *et al*., 2021] [32, 67].

Conclusion

From the above study we can conclude that, *Utricularia* is an intriguing genus which is now in focus of many researches related to phylogeny, taxonomy, trap biochemistry and many more. Largest portion of the genus ~84% comprises of terrestrial species such as lithophytes or epiphytes and ~16% are free floating aquatics. Recent studies have shown that, some species of *Utricularia* and *Genlisea* have the smallest genome size in angiosperms. Three lineages in *Utricularia* genus have evolved independently which have genome size of less than 100 Mbp. Many new species are also been identified all around the world. Different species of *Utricularia* have phytoremediation and medicinal properties. Still with all the development and research, many questions remain unsolved about its genome and trap pray interaction.

Acknowledgement

The authors are highly grateful to the authorities of Bharat Institute of Science and Technology (BIST), Bhubaneswar, Odisha for providing all the facilities to carry out the current study.

Conflict of Interest

Authors have no conflict of interest.

Reference

1. Adamec L. Functional characteristics of traps of aquatic carnivorous *Utricularia* species. Aquatic Botany. 2011;95:226-233. doi:10.1016/j.aquabot.2011.07.001
2. Adamec L, Sirová D, Vrba J, Rejmánková E. Enzyme production in the traps of aquatic *Utricularia* species. Biologia (Bratislava). 2010;65:273-278. doi:10.2478/s11756-010-0002-1
3. Albert VA, Jobson RW, Michael TP, Taylor DJ. The carnivorous bladderwort (*Utricularia*, Lentibulariaceae): a system inflates. Journal of Experimental Botany. 2010;61:5-9. doi:10.1093/jxb/erp349
4. Ali H, Khan E, Sajad MA. Phytoremediation of heavy metals: concepts and applications. Chemosphere. 2013;91:869-881.
5. Augustynowicz J, Grosicki M, Hanus-Fajerska E, Lekka M, Waloszek A, Kołoczek H. Chromium (VI) bioremediation by aquatic macrophyte *Callitriche cophocarpa* Sendtn. Chemosphere. 2010;79:1077-1083.
6. Augustynowicz J, Łukowicz K, Tokarz K, Płachno BJ. Potential for chromium (VI) bioremediation by the aquatic carnivorous plant *Utricularia gibba* L. (Lentibulariaceae). Environmental Science and Pollution Research. 2015;22:9742-9748. doi:10.1007/s11356-015-4151-1
7. Babović N, Drazić G, Djordjević A, Mihailović N. Heavy and toxic metal accumulation in six macrophyte species from fish pond Ečka, Republic of Serbia. In: Proceedings of BALWOIS; 2010 May 25-29; Ohrid, Republic of Macedonia. p. 1-6.

8. Bauer U, Di Giusto B, Skepper J, Grafe TU, Federle W. With a flick of the lid: a novel trapping mechanism in *Nepenthes gracilis* pitcher plants. PLoS ONE. 2012;7:e38951. doi:10.1371/journal.pone.0038951
9. Bell CD, Soltis DE, Soltis PS. The age and diversification of the angiosperms re-revisited. American Journal of Botany. 2010;97:1296-1303. doi:10.3732/ajb.0900346
10. Bohn HF, Federle W. Insect aquaplaning: *Nepenthes* pitcher plants capture prey with the peristome, a fully wettable water-lubricated anisotropic surface. Proceedings of the National Academy of Sciences USA. 2004;101:14138-14143. doi:10.1073/pnas.0405885101
11. Borišev M, Pajević S, Stanković Ž, Krstić B. Macrophytes as indicators and potential remediators in aquatic ecosystems: a case study. Large Rivers. 2008;18(1-2):107-115. doi:10.1127/lr/18/2008/107
12. Choosawad D, Leggat U, Dechsukhum C, Phongdra A, Chotigeat W. Anti-tumor activities of fucoidan from the aquatic plant *Utricularia aurea* Lour. Songklanakarin Journal of Science and Technology. 2005;27(3):799-807.
13. DeChaine EG. A bridge or a barrier? Bernicia's influence on the distribution and diversity of tundra plants. Plant Ecology and Diversity. 2008;1:197-207. doi:10.1080/17550870802328660
14. Deka N, Devi N. Aquatic angiosperms of BTC area, Assam, with reference to their traditional uses. Asian Journal of Plant Science and Research. 2015;5(5):9-13.
15. Divakar MC, John J, Vyshnavidevi P, Anisha AS, Govindum V. Herbal remedies of Madayipara hillock tribals in Kannur district, Kerala, India. Journal of Medicinal Plants Studies. 2013;1(6):34-42.
16. Eriksson O, Kainulainen K. The evolutionary ecology of dust seeds. Perspectives in Plant Ecology, Evolution and Systematics. 2011;13:73-87. doi:10.1016/j.ppees.2011.02.002
17. Fleischmann A. Taxonomic *Utricularia* news. Carnivorous Plant Newsletter. 2015;44:13-16.
18. Fleischmann A. The new *Utricularia* species described since Peter Taylor's monograph. Carnivorous Plant Newsletter. 2012;42:67-76.
19. Fleischmann A, Michael TP, Rivadavia F, Sousa A, Wang W, Temsch EM, *et al*. Evolution of genome size and chromosome number in the carnivorous plant genus *Genlisea* (Lentibulariaceae), with a new estimate of the minimum genome size in angiosperms. Annals of Botany. 2014;114:1651-1663. doi:10.1093/aob/mcu189
20. Forterre Y, Skotheim JM, Dumais J, Mahadevan L. How the Venus flytrap snaps. Nature. 2005;433:421-425. doi:10.1038/nature03185
21. Greilhuber J, Borsch T, Müller K, Worberg A, Porembski S, Barthlott W. Smallest angiosperm genomes found in Lentibulariaceae, with chromosomes of bacterial size. Plant Biology. 2006;8:770-777. doi:10.1055/s-2006-924101
22. Guisande C, Granado-Lorencio C, Andrade-Sossa C, Duque SR. Bladderworts. Functional Plant Science and Biotechnology. 2007;1:58-68.
23. Harms S, Johansson F. The influence of prey behaviour on prey selection of the carnivorous plant *Utricularia vulgaris*. Hydrobiologia. 2000;427:113-120. doi:10.1023/A:1003961614595
24. Ibarra-Laclette E, Lyons E, Hernández-Guzmán G, Pérez-Torres CA, Carretero-Paulet L, Chang TH, *et al*. Architecture and evolution of a minute plant genome. Nature. 2013;498:94-98. doi:10.1038/nature12132
25. Janarthanam MK, Henry AN. Bladderworts of India. Flora of India, Series 4. Calcutta: Botanical Survey of India; 1992.
26. Jobson RW, Albert VA. Molecular rates parallel diversification contrasts between carnivorous plant sister lineages. Cladistics. 2002;18:127-136. doi:10.1006/clad.2001.0187
27. Jobson RW. Five new species of *Utricularia* (Lentibulariaceae) from Australia. Telopea. 2013;15:127-142. doi:10.7751/telopea2013017
28. Jobson RW. *Utricularia corneliana* R.W. Jobson (Lentibulariaceae), a new species from North Kennedy district of Queensland. Austrobaileya. 2012;8:601-607.
29. Jobson RW, Baleeiro PC, Barrett MD. Six new species of *Utricularia* (Lentibulariaceae) from Northern Australia. Telopea. 2018;21:57-77. doi:10.7751/telopea12630
30. Jobson RW, Playford J, Cameron KM, Albert VA. Molecular phylogenetics of Lentibulariaceae inferred from plastid rps16 intron and trnL-F DNA sequences: implications for character evolution and biogeography. Systematic Botany. 2003;28:157-171. doi:10.1043/0363-6445-28.1.157
31. Król E, Płachno BJ, Adamec L, Stolarz M, Dziubińska H, Trebacz K. Quite a few reasons for calling carnivores "the most wonderful plants in the world". Annals of Botany. 2012;109:47-64. doi:10.1093/aob/mcr249
32. Kumar S, Devi P, Singh R. Traditional knowledge of medicinal plants among indigenous communities. Indian Journal of Traditional Knowledge. 2017;16(2):234-240.
33. Kumar S, *et al*. *Utricularia sunilii* (Lentibulariaceae), a striking new species from southern Western Ghats, Kerala, India. Phytotaxa. 2018;371(2):140-144. doi:10.11646/phytotaxa.371.2.9
34. Kumar S, Thorat SS, Mondal G, Singh PD, Labala RK, Singh LA, *et al*. Morphological characterization and ecology of *Utricularia aurea* Lour. Advances in Plants & Agriculture Research. 2019;9(1):61-63. doi:10.15406/apar.2019.09.00412
35. Liu J, Dong Y, Xu H, Wang D, Xu J. Accumulation of Cd, Pb and Zn by 19 wetland plant species in constructed wetland. Journal of Hazardous Materials. 2007;147(3):947-953. doi:10.1016/j.jhazmat.2007.01.125
36. Lobreau-Callen D, Jérémie J, Suarez-Cervera M. Morphologie et ultrastructure du pollen dans le genre *Utricularia* L. (Lentibulariaceae). Canadian Journal of Botany. 1999;77:744-767. doi:10.1139/cjb-77-5-744
37. Lonnig WE, Becker HA. Carnivorous Plants. Nature Encyclopedia of Life Sciences. 2004;1-7. doi:10.1038/npg.els.0003818
38. Malec P, Myśliwa-Kurdziel B, Prasad MNV, Waloszek A, Strzałka K. Role of aquatic macrophytes in biogeochemical cycling of heavy metals, relevance to soil-sediment continuum detoxification and ecosystem health. In: Sherameti I, Varma A, editors. Detoxification of heavy metals. Berlin: Springer; 2011. p. 345-368.

39. McPherson S. Carnivorous Plants and Their Habitats. Poole, Dorset, England: Redfern Natural History Productions; 2010.
40. Menezes CG, Gasparino EC, Baleeiro PC, Miranda VFO. Seed morphology of bladderworts: a survey on *Utricularia* sect. *Foliosa* and sect. *Psyllosperma* (Lentibulariaceae) with taxonomic implications. Phytotaxa. 2014;167:173-182.
41. Mette N, Wilbert N, Barthlott W. Food composition of aquatic bladderworts (*Utricularia*, Lentibulariaceae) in various habitats. Beiträge zur Biologie der Pflanzen. 2000;72:1-13.
42. Müller K, Borsch T. Phylogenetics of *Utricularia* (Lentibulariaceae) and molecular evolution of the trnK intron in a lineage with high substitutional rates. Plant Systematics and Evolution. 2005;250:39-67.
43. Müller KF, Borsch T, Legendre L, Porembski S, Barthlott W. Recent progress in understanding the evolution of carnivorous Lentibulariaceae (Lamiales). Plant Biology. 2006;8:748-757. doi:10.1055/s-2006-924706
44. Nouri M, Haddioui AEM. Assessment of metals contamination and ecological risk in Ait Ammar abandoned iron mine soil, Morocco. Ekológia (Bratislava). 2016;35(1):32-49. doi:10.1515/eko-2016-0003
45. O'Dea A, Lessios HA, Coates AG, *et al*. Formation of the Isthmus of Panama. Science Advances. 2016;2:e1600883. doi:10.1126/sciadv.1600883
46. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, *et al*. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ. 2021;372:n71. doi:10.1136/bmj.n71
47. Płachno BJ, Świątek P. Unusual embryo structure in viviparous *Utricularia nelumbiulfolia*, with remarks on embryo evolution in genus *Utricularia*. Protoplasma. 2010;239(1):69-80. doi:10.1007/s00709-009-0084-1
48. Płachno BJ, Adamec L, Kozieradzka-Kiszkurno M, Świątek P, Kamińska I. Cytochemical and ultrastructural aspects of aquatic carnivorous plant turions. Protoplasma. 2014;251:7-12. doi:10.1007/s00709-014-0646-8
49. Poppinga S, *et al*. Biomechanical analysis of prey capture in the carnivorous Southern bladderwort (*Utricularia australis*). Scientific Reports. 2017;7:1776. doi:10.1038/s41598-017-01954-3
50. Poppinga S, Joyeux M. Different mechanics of snap-trapping in the two closely related carnivorous plants *Dionaea muscipula* and *Aldrovanda vesiculosa*. Physical Review E. 2011;84:041928. doi:10.1103/PhysRevE.84.041928
51. Poppinga S, *et al*. Catapulting tentacles in a sticky carnivorous plant. PLoS ONE. 2012;7:e45735. doi:10.1371/journal.pone.0045735
52. Poppinga S, Masselter T, Speck T. Faster than their prey: new insights into the rapid movements of active carnivorous plant traps. BioEssays. 2013;35:649-657. doi:10.1002/bies.201200175
53. Poppinga S, Weißkopf C, Westermeier A, Masselter T, Speck T. Fastest predators in the plant kingdom: functional morphology and biomechanics of suction traps found in the largest genus of carnivorous plants. AoB Plants. 2016;8:plv140. doi:10.1093/aobpla/plv140
54. Prasad MNV, Greger M, Aravind P. Biogeochemical cycling of trace elements by aquatic and wetland plants: relevance to phytoremediation. In: Prasad MNV, Sajwan KS, Naidu R, editors. Trace elements in the environment, biogeochemistry, biotechnology, and bioremediation. Florida: CRC Press; 2006. p. 443-474.
55. Rahman MO. A taxonomic account of *Utricularia* Linn. from Bangladesh. Bangladesh Journal of Plant Taxonomy. 2005;12(2):63-70. doi:10.3329/bjpt.v12i2.603
56. Rai PK. Heavy metal phytoremediation from aquatic ecosystems with special reference to macrophytes. Critical Reviews in Environmental Science and Technology. 2009;39(9):697-753. doi:10.1080/10643380801910058
57. Raven JA. Rubisco: Still the most abundant protein of Earth? New Phytologist. 2013;198:1-3. doi:10.1111/nph.12197
58. Reut MS, Jobson RW. A phylogenetic study of subgenus *Polypompholyx*: a parallel radiation of *Utricularia* (Lentibulariaceae) throughout Australasia. Australian Systematic Botany. 2010;23:152-161.
59. Roccia A, Gluch O, Lampard S, Robinson A, Fleischmann A, McPherson S, *et al*. *Pinguicula* of the Temperate North. Dorset: Redfern Natural History Productions; 2016.
60. Roques AR, Jérémie J. Biodiversity in the genus *Utricularia* (Lentibulariaceae). Acta Botanica Gallica. 2005;152(2):177-186. doi:10.1080/12538078.2005.10515468
61. Rutishauser R. Evolution of unusual morphologies in Lentibulariaceae (bladderworts and allies) and Podostemaceae (river-weeds): a pictorial report at the interface of developmental biology and morphological diversification. Annals of Botany. 2016;117:811-832. doi:10.1093/aob/mcv172
62. Sasago A, Sibaoka T. Water extrusion in the trap bladders of *Utricularia vulgaris*. I. A possible pathway of water across the bladder wall. Botanical Magazine Tokyo. 1985a;98:55-66. doi:10.1007/BF02488906
63. Sasago A, Sibaoka T. Water extrusion in the trap bladders of *Utricularia vulgaris*. II. A possible mechanism of water outflow. Botanical Magazine Tokyo. 1985b;98:113-124. doi:10.1007/BF02488791
64. Shivakumar VS, Appelhans MS, Johnson G, Carlsen M, Zimmer EA. Analysis of whole chloroplast genomes from the genera of the Clauseneae, the Curry tribe (Rutaceae, Citrus family). Molecular Phylogenetics and Evolution. 2016;doi:10.1016/j.ympev.2016.12.015
65. Silva SR, Diaz YCA, Penha HA, Pinheiro DG, Fernandes C, Miranda VFO, Michael TP, Varani AM. The chloroplast genome of *Utricularia reniformis* sheds light on the evolution of the ndh gene complex of terrestrial carnivorous plants from the Lentibulariaceae family. PLoS ONE. 2016;doi:10.1371/journal.pone.0165176
66. Silva SR, Gibson R, Adamec L, Domínguez Y, Miranda VFO. Molecular phylogeny of bladderworts: a wide approach of *Utricularia* (Lentibulariaceae) species relationships based on six plastidial and nuclear DNA sequences. Molecular Phylogenetics and Evolution. 2018;118:244-264. doi:10.1016/j.ympev.2017.10.010

67. Singh P, Sharma R, Gupta M. Antioxidant and antimicrobial properties of *Utricularia* species: a review. Pharmacognosy Reviews. 2021;15(29):32-40.
68. Maksimović T, Rončević S, Kukavica B. *Utricularia vulgaris* L. and *Salvinia natans* (L.) All. heavy metal (Fe, Mn, Cu, Zn and Pb) bioaccumulation specificity in the area of Bardača fishpond. Ekológia (Bratislava). 2019;38(3):201-213.
69. Taylor P. The genus *Utricularia*: a taxonomic monograph. Kew Bulletin Additional Series XIV. London; 1989.
70. Veleba A, Bureš P, Adamec L, Šmarda P, Lipnerová I, Horová L. Genome size and genomic GC content evolution in the miniature genome-sized family. New Phytologist. 2014;203:22-28.
71. Albert VA, Jobson RW, Michael TP, Taylor DJ. The carnivorous bladderwort (*Utricularia*, Lentibulariaceae): a system inflates. Journal of Experimental Botany. 2010;61(1):5-9. doi:10.1093/jxb/erp349
72. Vincent O, Weißkopf C, Poppinga S, Masselter T, Speck T, Joyeux M, Quilliet C, Marmottant P. Ultra-fast underwater suction traps. Proceedings of the Royal Society B. 2011;278:2909-2914. doi:10.1098/rspb.2010.2292
73. Volkov AG, Forde-Tuckett V, Volkova MI, Markin VS. Morphing structures of the *Dionaea muscipula* Ellis during the trap opening and closing. Plant Signaling & Behavior. 2014;9:e27793-1-7.
74. Westermeier AS, Fleischmann A, Müller K, Schäferhoff B, Rubach C, Speck T, Poppinga S. Trap diversity and character evolution in carnivorous bladderworts (*Utricularia*, Lentibulariaceae). Scientific Reports. 2017;7:12052. doi:10.1038/s41598-017-12324-4
75. Yadav S, Sardesai MM, Gaikwad SP. A new species of *Utricularia* L. (Lentibulariaceae) from the Western Ghats, India. Rheedea. 2005;15(1):71-73.